own spot in this chapter. Make sure to check out *Programming Phoenix LiveView [TD24]* by Bruce Tate and Sophie DeBenedetto. It's the best book to get started with LiveView!

If you're building a LiveView app, you likely won't have a traditional HTTP-based API. But if you're not using LiveView, consider using GraphQL. Let's talk about that next.

GraphQL with Absinthe

GraphQL is a popular technology to build APIs with. It replaces traditional REST-based APIs with a completely different approach to API development. It can be a bit complex to get started with, but it's a fast and effective approach for API development. If you do decide to build a GraphQL API, the quality of your server implementation is important. Luckily, Elixir has one of the best implementations out there.

Absinthe[7] is the leading server implementation of GraphQL in Elixir. It provides all of the basics that you'd expect but also includes more advanced features such as subscriptions, full middleware control, and Relay support.

One of the hardest things to get right with a GraphQL API is batched resolution—you want to grab the data you need in the fewest number of queries possible. Absinthe has multiple approaches to solving this problem—arguably in a better way than other languages provide.

GraphQL is a paradigm shift from traditional ways of building APIs, but in my experience, it's worth it. The book *Craft GraphQL APIs in Elixir with Absinthe [WW18]* by Bruce Williams and Ben Wilson walks you through how to build your GraphQL with Absinthe, directly from the creators of the Absinthe framework.

Machine Learning with Nx and Bumblebee

Machine learning and artificial intelligence have been hot topics in recent years. Python is the most important language in the machine learning community, but Elixir is working on becoming a contender. Machine learning is vastly different than classical programming though, and it requires new ways of executing code.

Nx (Numerical Elixir)[8] was developed by Sean Moriarity and José Valim to solve the problem of machine learning in Elixir. Nx provides primitives that

7. https://hexdocs.pm/absinthe/overview.html
8. https://hexdocs.pm/nx/Nx.html